

Multi-Band Photometry and Spatial Analysis of the Globular Cluster NGC 5053

Probing Tidal Disruption via SExtractor and Gaia DR3

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ABSTRACT

Context. The globular cluster NGC 5053 is a metal-poor, low-density system in the outer Galactic halo. Its origin and dynamical evolution remain a subject of active debate, with growing evidence suggesting it may have been accreted from a disrupted satellite galaxy and is currently undergoing tidal stripping.

Aims. We aim to determine the physical parameters of NGC 5053 (age, metallicity, and distance) using multi-band optical and near-infrared photometry, and to investigate its spatial distribution for observational signatures of tidal interactions.

Methods. We perform photometry on g , r , and z band images from the Legacy Survey DR10. To overcome near-infrared observational challenges, z -band photometry is extracted using SExtractor in dual-image mode, anchored to the r -band detections. The stellar sample is purified via morphological classification, kinematic cross-matching with the Gaia DR3 catalog, and photometric error cuts. Cluster parameters are derived by fitting BaSTI theoretical isochrones to the extended $g - r$ and $g - z$ Color-Magnitude Diagrams (CMDs). Additionally, a Kernel Density Estimation (KDE) contour map is constructed to analyze spatial morphology.

Results. The best-fitting isochrones across both color planes consistently yield a low metallicity of $[M/H] = -1.597$ and a distance of approximately 16.37 kpc, with an age spanning 12.5 to 13 Gyr. The extended color baseline successfully separates evolutionary sequences and reveals a distinct Blue Straggler Star population. Furthermore, the spatial density map reveals a significant departure from spherical symmetry, exhibiting a pronounced elongation along the Right Ascension axis.

Conclusions. The consistent physical parameters derived across optical and near-infrared regimes firmly establish NGC 5053 as an ancient, metal-poor halo cluster. The distinct spatial asymmetry provides direct observational evidence of tidal tails, which, combined with the dynamically-driven presence of Blue Stragglers, strongly supports the hypothesis that the cluster is undergoing severe gravitational disruption by the Milky Way, consistent with an extragalactic origin.

Key words. galactic halo – globular clusters – NGC 5053 – photometry – kinematics

1. Introduction

1.1. Context

Beyond the characteristics of its stellar populations, the origin and dynamical evolution of NGC 5053 represent a fundamental aspect for understanding its nature. In the literature, there is an active debate regarding whether this cluster formed *in situ* within the Milky Way or if it was captured from a satellite galaxy, specifically the Sagittarius dwarf spheroidal galaxy (Sgr dSph) (Law & Majewski 2010). While recent orbital models in the outer halo strongly reinforce the hypothesis of an extragalactic capture origin (Gontcharov et al. 2024), some chemical abundance studies challenge a direct association with the Sgr stream, leaving the exact progenitor an open question (Tang et al. 2018). Furthermore, NGC 5053 is a cluster with a very low central density, making it highly susceptible to gravitational perturbations from the Milky Way or its satellites (Gontcharov et al. 2024).

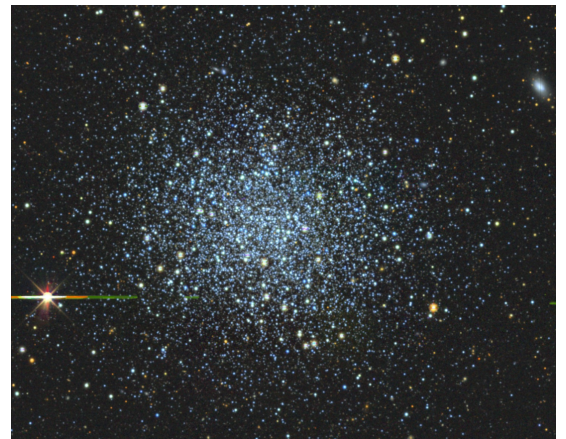


Fig. 1: Optical image of the globular cluster NGC 5053 from the Legacy Survey DR10.

1.2. Data Selection

For this study, astronomical images of the globular cluster NGC 5053 were utilized, obtained from the *Legacy Survey DR10*

(Dey et al. 2019) in the g , r , and z photometric filters. Additionally, to perform the astrometric cleaning of the sample, the *Gaia DR3* catalog (Gaia Collaboration et al. 2023) was employed, providing high-precision astrometric and kinematic measure-

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ments (proper motions) essential for distinguishing cluster members from field stars. For the astrophysical interpretation and determination of physical parameters, theoretical isochrones were extracted from the BaSTI (*Bag of Stellar Tracks and Isochrones*) database (Pietrinferni et al. 2021).

Table 1: Characteristics of the photometric filters utilized from the Legacy Survey DR10.

Filter	λ_{eff} (Å)	Spectral Region
<i>g</i>	4720	Optical (blue-green)
<i>r</i>	6415	Optical (red)
<i>z</i>	9260	Near-infrared

2. Methodology

After performing photometry with SExtractor (Bertin & Arnouts 1996) on NGC 5053 using the *g*, *r*, and *z* filter images, the morphological, positional, and photometric parameters detailed in Table 2 were extracted.

Table 2: SExtractor output parameters extracted for the analysis.

Parameter	Description
NUMBER	Running object number
FLUX_APER	Flux vector within fixed circular aperture(s)
FLUXERR_APER	RMS error vector for aperture flux(es)
MAG_APER	Fixed aperture magnitude vector
MAGERR_APER	RMS error vector for fixed aperture mag.
BACKGROUND	Background at centroid position
X_IMAGE	Object position along x
Y_IMAGE	Object position along y
ALPHA_J2000	Right ascension of barycenter (J2000)
DELTA_J2000	Declination of barycenter (J2000)
A_WORLD	Profile RMS along major axis
B_WORLD	Profile RMS along minor axis
THETA_WORLD	Position angle (CCW/world-x)
FLAGS	Extraction flags
FWHM_IMAGE	FWHM assuming a gaussian core
ELLIPTICITY	$1 - B_IMAGE/A_IMAGE$
CLASS_STAR	S/G classifier output

Since SExtractor does not inherently discriminate between field stars and cluster members, it performs photometry on all objects detected in the .fits file. Consequently, the initial catalog is contaminated by background objects, foreground stars, and unresolved sources (such as galaxies or nebulae). To clean the sample and isolate the stellar population of NGC 5053, three sequential selection criteria were applied:

First, for morphological discrimination, the *CLASS_STAR* parameter from SExtractor was used, which classifies objects on a continuous scale between 0 and 1. Values close to 1 indicate a profile compatible with a point source (star), while values close to 0 indicate extended sources like galaxies. The distribution of this parameter exhibits a bimodal behavior (Figure 2), with a valley populated by noise sources and unresolved objects. By comparing this with the apparent magnitude in Figure 3, a clear grouping of stars is observed, which shifts away from 1 at fainter magnitudes. From a magnitude of ~ 21 , the low surface brightness makes it difficult to resolve sources against the background noise, causing them to fall completely into intermediate classifications in the $\sim 22 - 23$ magnitude range. To avoid discarding intrinsically faint real stars or those affected by crowding noise, a

cutoff threshold was established at $CLASS_STAR \geq 0.7$. To ensure higher completeness, it was required that the average value reported by SExtractor between the *g* and *r* filters met this condition.

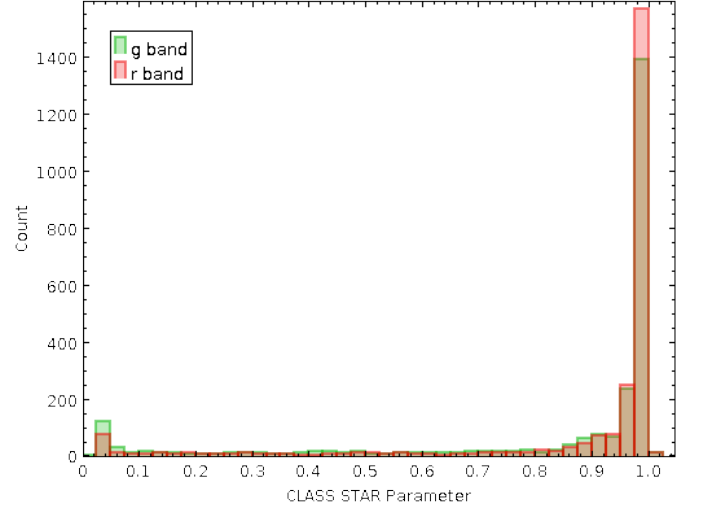


Fig. 2: Distribution of the *CLASS_STAR* parameter for the detected sources. The bimodal shape separates point sources (near 1.0) from extended objects (near 0.0).

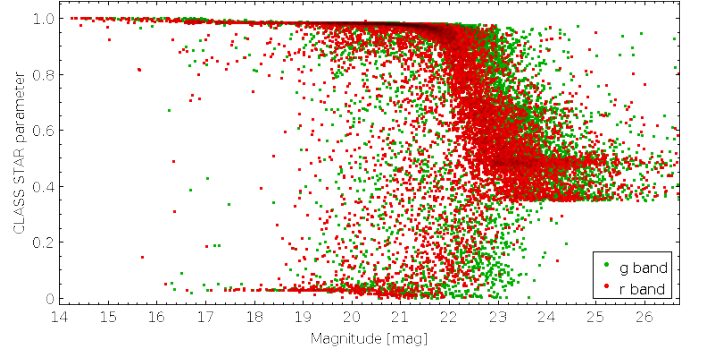


Fig. 3: *CLASS_STAR* index as a function of apparent magnitude. Fainter objects suffer from increased noise, causing their classification to diverge from 1.0. A conservative threshold of $CLASS_STAR \geq 0.7$ was adopted.

Second, to filter stellar field contaminants (foreground or background), kinematics were utilized. Stars belonging to a globular cluster move together as a gravitationally bound system in the sky, whereas field objects possess random, unbound motions. Using TOPCAT, the SExtractor data were cross-matched with the *Gaia* DR3 catalog, employing spatial coordinates as the matching key (Nikitha et al. 2022; Gontcharov et al. 2024). This allowed the incorporation of proper motion columns and the manual selection of the overdensity corresponding to the cluster members in the phase space, as detailed in Figure 4.

Third, a photometric quality criterion was applied. As evidenced in Figure 5, fainter objects present a significantly larger photometric error dispersion. Photometry was defined as reliable if the error reported by SExtractor was 0.6 magnitudes or less, a limit corresponding to the "elbow" of the error growth curve.

To observationally investigate the signatures of dynamical interaction with our catalog data, a 2D smoothed density map

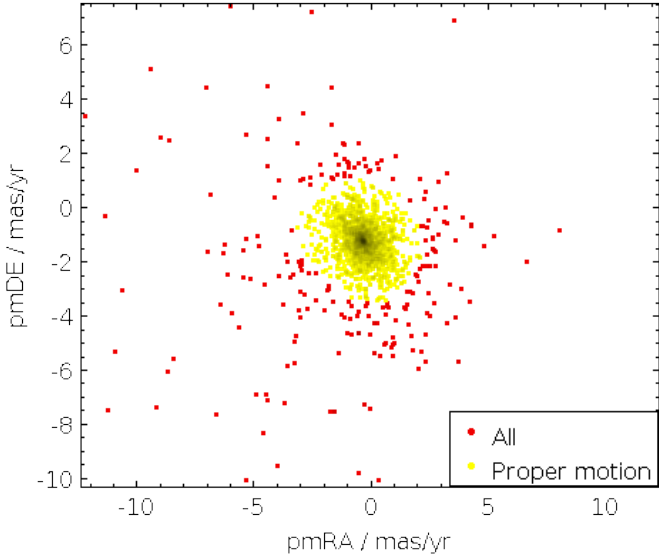


Fig. 4: Vector Point Diagram (VPD) of the proper motions from Gaia DR3. The distinct dense clustering of points represents the kinematically bound members of NGC 5053.

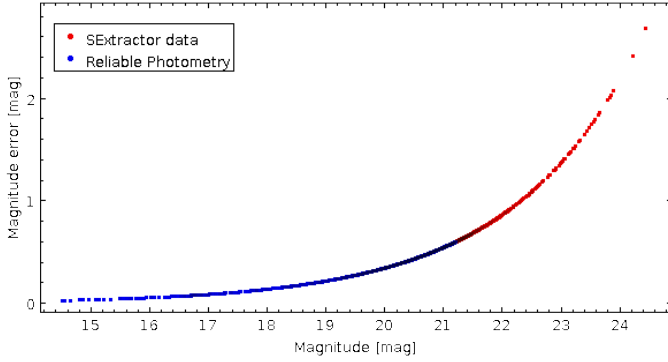


Fig. 5: Photometric error as a function of magnitude. A reliability cut of ≤ 0.6 mag was applied to ensure the robustness of the resulting Color-Magnitude Diagram.

(contours) of the cluster’s spatial distribution was constructed. Using TOPCAT, exclusively definitive cluster members (previously purified via *Gaia* DR3 kinematics and photometric error cuts) were isolated. A Kernel Density Estimation (KDE) algorithm was applied over these spatial coordinates using a logarithmic scale. This analytical technique suppresses individual stellar noise and visually enhances the low-density peripheral structures that remain invisible in a traditional scatter plot.

To extend the photometric analysis into the near-infrared, z -band imagery was incorporated. However, due to the inherently higher sky background and airglow typical of z -band observations, morphological classification using a single-image approach yielded a high contamination rate, frequently misclassifying stellar sources as extended objects. To robustly resolve this, photometry was extracted using SExtractor in dual-image mode. The r -band image, possessing a superior signal-to-noise ratio and stable seeing, was utilized as the detection reference to define apertures, determine centroids, and compute the CLASS_STAR morphological parameter. Subsequently, the z -band image was used strictly for measurement within these pre-defined apertures. This approach ensured consistent cross-band

photometry while preserving the accurate star-galaxy separation inherited from the optical bands.

3. Results

The definitive stellar population of the cluster was composed exclusively of objects meeting all three criteria: an average morphological stellar classification ≥ 0.7 , kinematic membership in the proper motion subset, and reliable photometry with an error ≤ 0.6 mag.

3.1. Spatial Distribution

The resulting spatial distribution of the definitive, filtered members is shown in Figure 6. The apparent void in the center is an artifact caused by severe stellar crowding, which prevents SExtractor from resolving individual point sources. However, the purified sample successfully visualizes the cluster’s extended structure and allows for accurate photometric analysis.

3.2. Physical Parameters: Age, Metallicity, and Distance

With the kinematic members isolated, the Color-Magnitude Diagram (CMD) was constructed using the g and r filters (Figure 7). The CMD clearly reveals the typical evolutionary sequences of an old globular cluster.

To determine the fundamental physical parameters of NGC 5053, a theoretical isochrone extracted from BaSTI was overlaid onto the observational CMD (Figure 8). The best visual fit was achieved with the parameters detailed in Table 3, highlighting a very low metallicity ($[M/H] = -1.597$) and an advanced age of 13 Gyr, consistent with galactic halo populations (Nikitha et al. 2022; Gontcharov et al. 2024).

Table 3: Physical parameters of the selected BaSTI isochrone.

Parameter	Value
Metallicity ($[M/H]$)	-1.597
Metal fraction (Z)	0.0003974
Helium abundance (Y)	0.247506
Age (t)	13 Gyr

To align the isochrone (which uses theoretical absolute magnitudes) with the observational data (apparent magnitudes), a vertical shift of $+16.1$ mag was necessary. This establishes the relationship:

$$M_g + 16.1 = m_g \quad (1)$$

Which implies that the apparent distance modulus (μ_g) for this cluster in the g filter is:

$$m_g - M_g = \mu_g = 16.1 \quad (2)$$

From this, the distance d (in parsecs) to the cluster estimated using SExtractor photometry and the theoretical isochrone is:

$$d = 10^{\frac{\mu_g + 5 - A_g}{5}} \quad (3)$$

Where A_g corresponds to the interstellar extinction in the g band. This extinction is derived from the reddening, which was

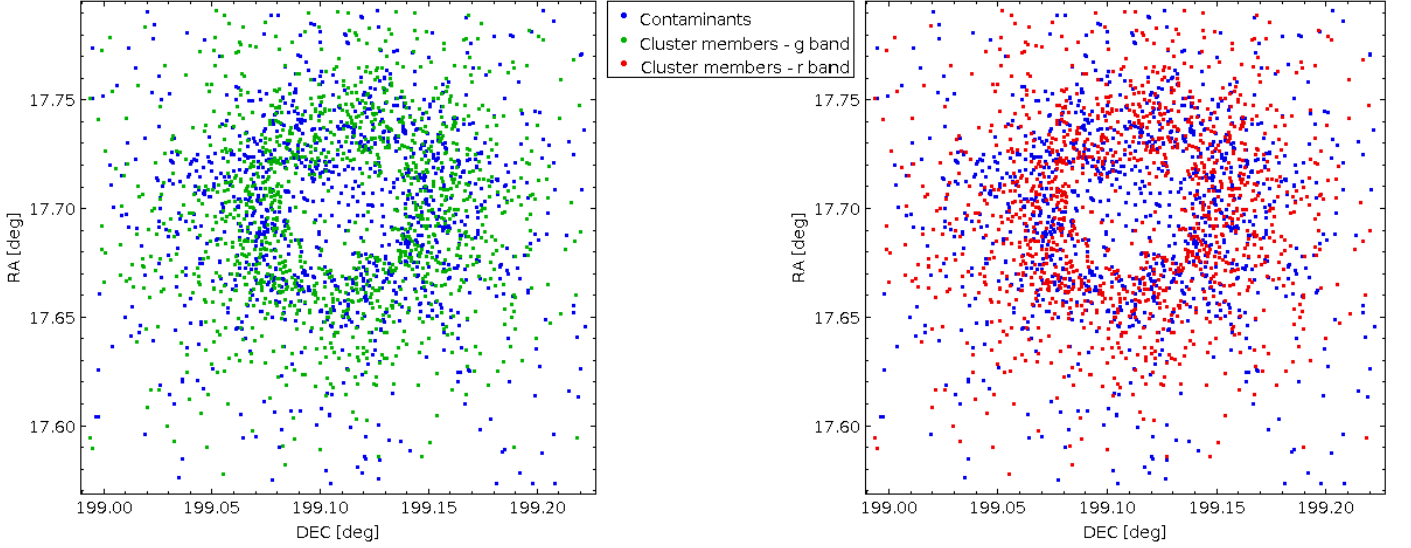


Fig. 6: Spatial distribution of the definitive, filtered members of NGC 5053.

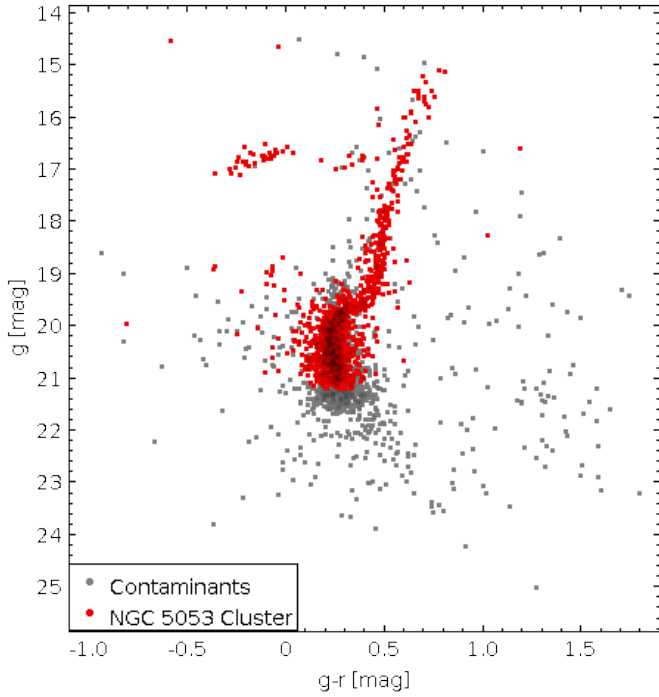


Fig. 7: Color-Magnitude Diagram (CMD) in the g and r bands for the filtered kinematic members of NGC 5053.

adjusted by shifting the isochrone along the abscissa axis by -0.009 magnitudes:

$$m_g - m_r - 0.009 \implies E(g-r) = 0.009 \quad (4)$$

Using the standard relation, for the g band, extinction and reddening are approximately related as:

$$A_g \approx 3.3 \times E(g-r) \approx 2.97 \times 10^{-2} \quad (5)$$

By substituting these values into the distance modulus equation, the physical distance is obtained:

$$d \approx 16370 \text{ pc} \quad (6)$$

3.3. Multi-Band Validation in the Near-Infrared

The incorporation of the z -band filter significantly extends the color baseline of the study, as demonstrated in Figure (9). By superimposing the $g-r$ and $g-z$ CMDs, a pronounced horizontal stretching of the evolutionary sequences is observed. This extended baseline provides greater sensitivity to the effective temperatures of cooler, lower-mass stars. Consequently, the steep Red Giant Branch (RGB) seen in the optical plane is resolved into a more extended and measurable slope, while foreground and background contaminants are more efficiently separated from the cluster's main sequence.

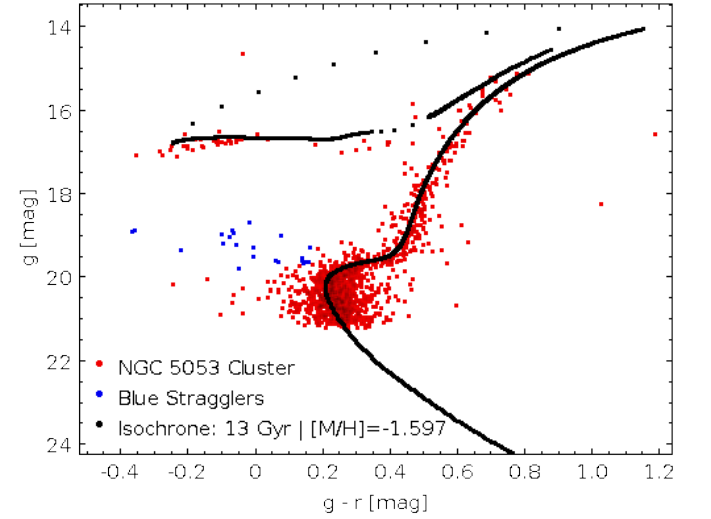


Fig. 8: Best-fitting BaSTI theoretical isochrone overlaid on the CMD, representing an ancient (13 Gyr), metal-poor stellar population. The blue points highlight the selected subset of Blue Straggler Stars (BSS), which distinctly deviate from the theoretical turn-off.

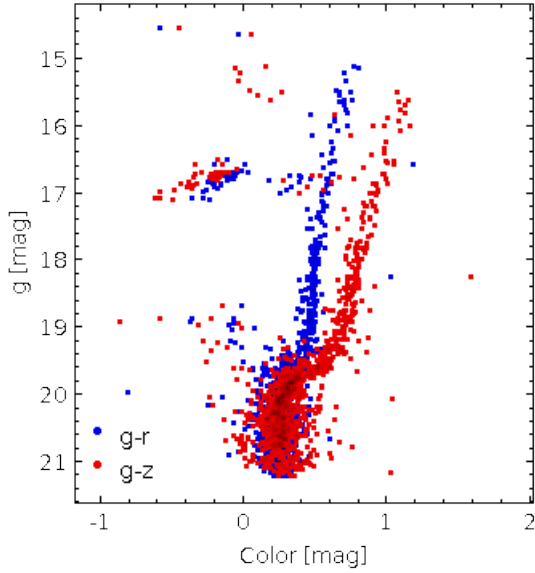


Fig. 9: Superimposed CMDs demonstrating the extended color sensitivity of the $g - z$ baseline compared to $g - r$.

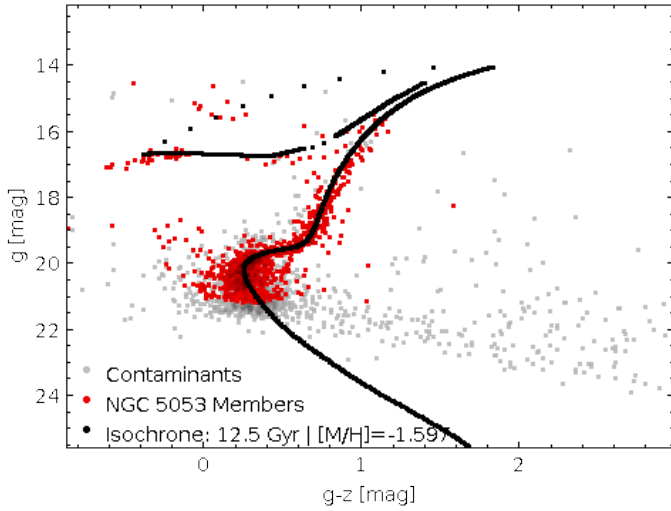


Fig. 10: The 12.5 Gyr BaSTI isochrone successfully reproducing the $g - z$ observational data, mapped with a distance modulus of 16.1 mag and a color shift of +0.03 mag.

Figure 10 presents the theoretical BaSTI isochrone fit applied to the $g - z$ plane. The best visual fit was achieved with an age of 12.5 Gyr, a distance modulus of $(m - M)_g = 16.1$ mag, and a minor color excess adjustment of $E(g - z) = 0.03$ mag.

3.4. The Blue Straggler Star Population

A notable feature in the color-magnitude diagram (highlighted in blue in Figure 8) is the distinct presence of a Blue Straggler Star (BSS) subpopulation. These objects were selected based on their position above and blueward of the main sequence turn-off (MSTO), spanning an approximate magnitude range of $17.5 \lesssim g \lesssim 19.0$ and a color range of $0.0 \lesssim g - r \lesssim 0.2$.

By significantly deviating from the 13 Gyr theoretical isochrone and appearing as a hotter, more massive extension of the main sequence, this anomalous population pro-

vides strong observational evidence for binary evolution or collisional processes. Despite its low density, NGC 5053 hosts a rich BSS population, which is highly consistent with the cluster's history of tidal disruption and dynamical interactions (Sarajedini & Milone 1995; Nikitha et al. 2022).

3.5. Tidal disruption evidence

The resulting KDE contour map (Figure 11) reveals crucial information regarding the morphological state of the cluster. In an isolated, gravitationally relaxed stellar system, isodensity contours exhibit perfect spherical symmetry. However, in the case of NGC 5053, it is observed that the outer regions completely lose this spherical symmetry, presenting a marked directional elongation along the Right Ascension axis and irregular contours.

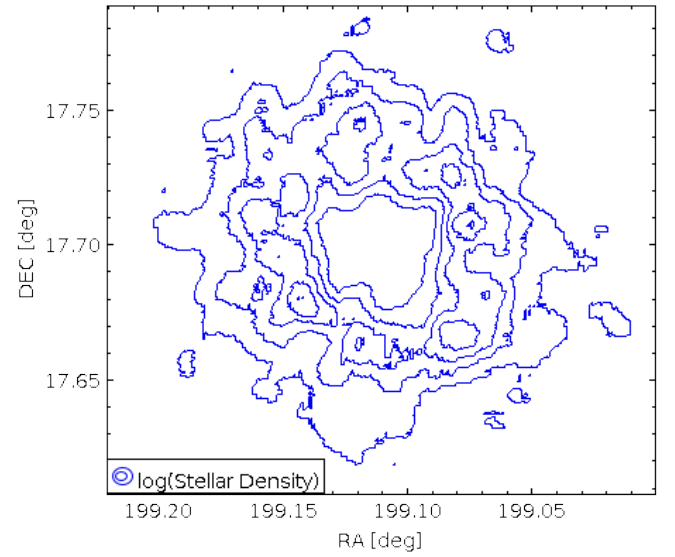


Fig. 11: Kernel Density Estimation (KDE) contour map of the filtered cluster members. The axes represent Right Ascension and Declination in degrees. The logarithmic density scale reveals a significant spatial elongation and non-spherical symmetry in the cluster's periphery.

4. Discussion

4.1. Methodological Limitations

The spatial analysis of the selected members in Figure 6 reveals an important observational phenomenon: there appears to be a stellar void in the center of the cluster in both filters. This effect does not represent a true lack of stars in the core, but rather is the product of methodological and instrumental limitations in the face of severe stellar crowding. In the innermost region, high surface density causes the brightness profiles of adjacent stars to overlap. Consequently, SExtractor fails to resolve individual sources (Bertin & Arnouts 1996), cataloging these light conglomerates with low *CLASS_STAR* values or assigning them photometric errors above the 0.6 mag threshold, causing them to be filtered out by our criteria. This limitation highlights the difficulty of performing traditional aperture photometry in dense cores without resorting to Point Spread Function (PSF) fitting techniques.

4.2. Validation of Physical Parameters

Despite these core limitations, the photometry obtained in the outer regions was of sufficient quality to define the evolutionary branches in the CMD and estimate the physical parameters. The obtained distance of ~ 16.37 kpc is consistent with contemporary literature estimates, which place NGC 5053 at distances on the order of 16.99 ± 0.56 kpc (Gontcharov et al. 2024) and 17.54 ± 0.23 kpc (Nikitha et al. 2022). Similarly, the 13 Gyr model is highly concordant with deep studies based on *Gaia* DR3 kinematics, which attribute ages between 12.5 and 12.7 Gyr to it (Nikitha et al. 2022; Gontcharov et al. 2024).

The independent isochrone fitting in the $g - z$ plane yields physical parameters that are remarkably consistent with this optical $g - r$ analysis. The required distance modulus ($+16.1$ mag) remains identical in both fits, firmly anchoring the spatial distance to NGC 5053. The slight variation in the derived age—12.5 Gyr in the near-infrared compared to 13.0 Gyr in the optical—is a well-documented effect in multi-band photometry. This minor discrepancy is often attributable to uncertainties in the theoretical modeling of cool star atmospheres and complex molecular opacities required to transform theoretical temperatures into near-infrared colors, rather than a physical contradiction (Gallart et al. 2005; Hidalgo et al. 2018). The applied color shift ($+0.03$ mag) accurately reflects the minimal reddening expected towards this high-galactic-latitude cluster. Ultimately, the successful inclusion of the z -band breaks potential age-metallicity degeneracies, validating the physical parameters derived from the optical bands and proving the robustness of the dual-image extraction method.

4.3. Dynamical History and Tidal Disruption evidence

Additionally, the age parameter and the very low metallicity ($[M/H] = -1.597$) point to NGC 5053 as an extremely ancient system, characteristic of the outer halo. It is important to note that NGC 5053 exhibits signatures of dynamical interaction. Its relatively sparse structure and low core density suggest it is highly susceptible to the tidal effects of the Milky Way's gravitational potential, interactions that result in the gradual loss of its lower-mass stars (Gontcharov et al. 2024). The spatial deformation observed in Figure 11 is the direct observational footprint of tidal tails, empirically demonstrating that the cluster is undergoing strong disruption and losing stars from its outer layers due to interaction with the galactic gravitational potential.

This scenario of dynamical perturbations and possible collisional evolution is also congruent with the presence of Blue Stragglers in the cluster, which typically form through physical collisions promoted by dense environments or past gravitational alterations (Nikitha et al. 2022). Future investigations could expand the spatial search radius using this same kinematic catalog to track the full extent of these tidal tails and further confirm its interacting nature.

In conclusion, the asymmetry evidenced in the spatial distribution of the photometric data, coupled with the low metallicity ($[M/H] = -1.597$) and the highly consistent age estimates derived from both optical and near-infrared isochrone fitting (13.0 and 12.5 Gyr, respectively), form a coherent physical picture. The validation across extended color baselines strongly anchors the cluster's fundamental parameters. These observational results not only confirm that NGC 5053 is a strongly perturbed system, but they are fully compatible with the cosmological scenario that the cluster is an accreted object of extragalactic origin (Law & Majewski 2010), even if its specific chemical link

to the Sagittarius dwarf galaxy remains under debate (Tang et al. 2018). Furthermore, this environment of tidal disruption is consistent with the collisional formation of the Blue Straggler population observed in the CMD (Sarajedini & Milone 1995).

5. Conclusions

The multi-band photometry of the globular cluster NGC 5053, successfully extracted using SExtractor—including a dual-image mode approach to overcome near-infrared observational challenges in the z -band—and strictly filtered using *Gaia* DR3 kinematics, has yielded an unpolluted catalog of cluster members. By fitting BaSTI theoretical isochrones to the constructed Color-Magnitude Diagrams across extended color baselines ($g - r$ and $g - z$), we robustly derived a distance of approximately 16.37 kpc, an age spanning 12.5 to 13 Gyr, and a metallicity of $[M/H] = -1.597$. The high degree of consistency between the optical and near-infrared evolutionary planes breaks potential age-metallicity degeneracies and firmly confirms the cluster's nature as an ancient, metal-poor system residing in the outer galactic halo.

Furthermore, spatial analysis using Kernel Density Estimation (KDE) contour maps revealed a significant departure from spherical symmetry in the cluster's periphery. The pronounced elongation along the Right Ascension axis provides direct observational evidence of tidal tails. This morphological distortion strongly supports the hypothesis that NGC 5053 is undergoing severe gravitational disruption by the Milky Way. Ultimately, these findings, coupled with the dynamically-driven presence of Blue Straggler Stars, are highly consistent with the scenario proposing that NGC 5053 is not indigenous to our galaxy, but rather an accreted remnant from a disrupted satellite system.

Data Availability

The astronomical images utilized in this study are publicly available through the DESI Legacy Imaging Surveys DR10 database. To ensure full reproducibility of the photometric extraction and spatial analysis, the resulting catalogs, subset masks, and all SExtractor configuration files (including dual-image mode parameters) are openly hosted in a public GitHub repository.

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